Energy-momentum tensor

Charge conservation implies continuity equ for the electric current
(4-scalar) (4-vector)

What corresponds to conservation of 4-momentum $\left(\frac{\varepsilon}{c}, \vec{p}\right)$?
$\Rightarrow$ the flow of 4-momentum must be 4-tensor
$\downarrow$
<u>energy-momentum tensor</u>

Start with a system of free particles (of one type):
it is described by $\vec{v}(t,\vec{r})$ and mass density $\rho_m(t,r) = \frac{dm}{dV}$
The 4-momentum of particles inside volume dV
is $u^\alpha dm$ $\left[u^\alpha = \gamma(c, \vec{v}) \right]$

$\Rightarrow$ 4-momentum density $p^\alpha = \frac{u^\alpha dm}{dV} = \rho_m u^\alpha$

$$p^0 = \rho_m \gamma c = \frac{1}{c}\left(\gamma \rho_m c^2\right) = \frac{W}{c} \quad \text{energy density}$$

$$p^i = \gamma \rho_m v^i \quad \text{momentum density}$$

$$\left(\text{Note } \gamma = \frac{1}{\sqrt{1 - \frac{\vec{v}^2(t,\vec{r})}{c^2}}} = \gamma(t,\vec{r}).\right)$$

$$\frac{\partial p^\alpha}{\partial t} = \frac{\partial \rho_m}{\partial t} u^\alpha + \rho_m \frac{\partial u^\alpha}{\partial t}$$

Note: $\rho_m(t,\vec{r}) = \sum_a m_a \delta(\vec{r} - \vec{r}_a(t))$

$$\frac{\partial \rho_m}{\partial t} = \sum_a m_a \underbrace{\frac{\partial}{\partial \vec{r}_a} \delta(\vec{r} - \vec{r}_a)}_{-\frac{\partial}{\partial \vec{r}} \delta(\vec{r} - \vec{r}_a)} \underbrace{\frac{\partial \vec{r}_a}{\partial t}}_{\vec{v}_a} = -\sum_a m_a \vec{v}_a \cdot \vec{\nabla} \delta(\vec{r} - \vec{r}_a)$$

$$= -\vec{\nabla} \sum_a m_a \vec{v}_a \delta(\vec{r} - \vec{r}_a) = -\frac{\partial}{\partial x^i}\left(\rho_m v^i\right) = -\frac{\partial}{\partial x^i}\left(\rho_m \frac{u^i}{\gamma}\right)$$

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Newton's law : $m \frac{du^\alpha}{d\tau} = q F^{\alpha\beta} u_\beta$

$$\Rightarrow \frac{m}{q} \frac{du^\alpha}{d\tau} = \frac{m}{q} \gamma \frac{du^\alpha}{dt} = F^{\alpha\beta} u_\beta$$

$$d\tau = \frac{dt}{\gamma}$$

But $\frac{m}{q} = \frac{\rho_m}{\rho}$ here we use $m_a = m \Rightarrow$ one type of particles

$$\Rightarrow \rho_m \frac{du^\alpha}{dt} = \rho F^{\alpha\beta} \frac{u_\beta}{\gamma} = F^{\alpha\beta} \underbrace{\left(\rho \frac{u_\beta}{\gamma}\right)}_{= j_\beta \text{ 4-current}} = F^{\alpha\beta} j_\beta$$

total derivative (while we need $\frac{\partial u^\alpha}{\partial t}$ = partial)

$$\frac{dA}{dt} = \frac{\partial A}{\partial t} + \frac{\partial A}{\partial \vec{r}} \cdot \frac{\partial \vec{r}}{\partial t} = \frac{\partial A}{\partial t} + (\vec{v} \cdot \vec{\nabla}) A$$

derivative for specific particle

derivative at specific point in space

since $A(t, \vec{r}(t))$ depends not only t but also on particles motion in space.

$$\Rightarrow \rho_m \frac{du^\alpha}{dt} = \rho_m \frac{\partial u^\alpha}{\partial t} + \rho_m v^i \frac{\partial u^\alpha}{\partial x^i} = F^{\alpha\beta} j_\beta$$

$$\Rightarrow \text{so} \quad \frac{\partial P^\alpha}{\partial t} = -u^\alpha \frac{\partial}{\partial x^i}\left(\rho_m \frac{u^i}{\gamma}\right) \underbrace{- \rho_m \overset{u^i/\gamma}{v^i} \frac{\partial u^\alpha}{\partial x^i} + F^{\alpha\beta} j_\beta}_{\rho_m \frac{\partial u^\alpha}{\partial t}}$$

$$= -\frac{\partial}{\partial x^i}\underbrace{\left(\frac{\rho_m}{\gamma} u^\alpha u^i\right)}_{\text{flux of } \alpha\text{-comp. of 4-momentum along } i\text{-th axis}} + \underbrace{F^{\alpha\beta} j_\beta}_{\text{change of 4-mom. due to EM forces}}$$

looks like continuity eqn

Notice $u^0 = c\gamma \Rightarrow \frac{\partial p^\alpha}{\partial t} = \frac{\partial}{\partial t}\left(\rho_m u^\alpha\right) = \frac{\partial}{\partial t}\left(\rho_m u^\alpha \frac{u^0}{c\gamma}\right) =$

$$= \frac{\partial}{c\partial t}\left(\frac{\rho_m}{\gamma} u^\alpha u^0\right) = \frac{\partial}{\partial x^0}\left(\frac{\rho_m}{\gamma} u^\alpha u^0\right)$$

$$\Rightarrow \frac{\partial p^\alpha}{\partial t} + \frac{\partial}{\partial x^i}\left(\frac{\rho_m}{\gamma} u^\alpha u^i\right) = \frac{\partial}{\partial x^\beta}\left(\frac{\rho_m}{\gamma} u^\alpha u^\beta\right) = \frac{\partial}{\partial x^\beta}\left(T^{\alpha\beta}\right)$$

$$T^{\alpha\beta}_{\text{particle}} = \frac{\rho_m}{\gamma} u^\alpha u^\beta = \rho_m \sqrt{1-\frac{v^2}{c^2}}\, u^\alpha u^\beta$$

energy-momentum or stress-energy tensor

(1) $$\frac{\partial}{\partial x^\beta}\left(T^{\alpha\beta}\right) = F^{\alpha\beta} j_\beta = F^{\alpha\delta} j_\delta$$

(summation index $\beta \to \delta$)

4-tensor $\Leftarrow$ 4-vector

Observe $T^{\alpha\beta} = T^{\beta\alpha} = \frac{\rho_m}{\gamma} u^\alpha u^\beta$

$$T^{\alpha 0} = T^{0\alpha} = \frac{\rho_m}{\gamma} u^\alpha u^0 = c\left(\rho_m u^\alpha\right) = c\, p^\alpha$$

4-momentum density

$$T^{00} = \frac{\rho_m}{\gamma} u^0 u^0 = \underbrace{c^2 \rho_m \gamma}_{W} = c\, p^0$$

$$W = \frac{\rho_m c^2}{\sqrt{1-v^2/c^2}} \quad \text{energy density}$$

$$\cancel{\frac{\partial}{\partial x^0}\left(T^{00}\right) = \frac{\partial}{\partial x^0}\left(T^{0\alpha}\right) = \frac{1}{c}\frac{\partial}{\partial t}}$$

$$\frac{\partial}{\partial t} W = \frac{\partial}{\partial t} T^{00} = c\frac{\partial}{\partial x^0} T^{00} = \frac{\partial}{\partial x^0}\left(c T^{00}\right) =$$

$$= -\frac{\partial}{\partial x^i}\left(c T^{0i}\right) + c F^{0\beta} j_\beta$$

$$F^{0\beta} = (0, -E_x/c, -E_y/c, -E_z/c) \quad j_\beta = (c\rho, -\vec{j})\,4$$

$$\frac{\partial W}{\partial t} = -c\,\frac{\partial T^{0i}}{\partial x^i} + cF^{0\beta} j_\beta = -c^2 \frac{\partial p^i}{\partial x^i} + \vec{j}\cdot\vec{E}$$

power (work/time) of electric forces acting on moving charges.

$$\Downarrow$$

$T^{0i} = T^{i0}$ = energy flux along i^{th} direction)

while $\dfrac{\partial p^i}{\partial t} = \dfrac{1}{c}\dfrac{\partial}{\partial t} T^{i0} = \dfrac{1}{c}\dfrac{\partial T^{0i}}{\partial t} = \dfrac{\partial}{\partial x^0}(T^{0i}) = \dfrac{\partial}{\partial x^0}(T^{i0})$

$$= \cancel{\frac{\partial}{\partial x^0}(T^{0i}) = -\frac{\partial}{\partial x^j} A^{ij}}$$

$$(\text{use } (1)) \quad = -\frac{\partial}{\partial x^j}(T^{ij}) + \underbrace{F^{i\beta} j_\beta}_{(\rho\vec{E} + \vec{j}\times\vec{B})^i}$$

; e.g. $F^{1\beta} j_\beta = \frac{E_x}{c} c\rho - B_z(-j_y) + B_y(-j_z) = \rho E_x + j_y B_z - j_z B_y = (\rho\vec{E} + \vec{j}\times\vec{B})_x$

$$\int_V dV \frac{\partial p^i}{\partial t} = \underbrace{\frac{\partial}{\partial t}\int_V dV\, p^i}_{\text{rate of change of total momentum of particles inside } V.} = -\oint_S da_j\, T^{ij} + \int_{dV} dV (\rho\vec{E} + \vec{j}\times\vec{B})^i$$

surface total force — bulk (Lorentz)

Surface force $-T^{ij} da_j = +dF^i$ is convective force due to the flow of momentum in/out the surface element $d\vec{a}$.

Often $T^{ij} = -\sigma^{ij}$, σ^{ij} = Maxwell's stress tensor.

Stress-energy tensor of EM field

In the absence of EM field

$$\frac{\partial}{\partial x^\beta}\left(T^{\alpha\beta}_{\text{particle}}\right) = 0 .$$

With EM present, we expect the total energy-momentum (particles + field) to be conserved too $\Rightarrow$

$$\underbrace{\frac{\partial}{\partial x^\beta}\left(T^{\alpha\beta}_{\text{part}}\right)}_{F^{\alpha\beta} j_\beta} + \frac{\partial}{\partial x^\beta}\left(T^{\alpha\beta}_{\text{field}}\right) = 0.$$

$$F^{\alpha\beta} j_\beta \Rightarrow \frac{\partial}{\partial x^\beta}\left(T^{\alpha\beta}_{\text{field}}\right) = -F^{\alpha\beta} j_\beta = -\left(\vec{j}\cdot\vec{E}, \rho\vec{E} + \vec{j}\times\vec{B}\right)$$

Expect $T^{\alpha\beta}_{\text{field}}$ to be determined by the field only

$\Rightarrow$ need to express particle current j_β via field

Gauss-Ampere $\frac{\partial F^{\alpha\beta}}{\partial x^\beta} = -\mu_0 j^\alpha \Rightarrow j^\beta = -\frac{1}{\mu_0}\frac{\partial F^{\beta\gamma}}{\partial x^\gamma}$

$$j_\beta = -\frac{1}{\mu_0}\frac{\partial F_\beta{}^\gamma}{\partial x^\gamma}$$

$$\Rightarrow \frac{\partial T^{\alpha\beta}}{\partial x^\beta} = \frac{1}{\mu_0}\underbrace{F^{\alpha\beta}\frac{\partial F_\beta{}^\mu}{\partial x^\mu}}$$

write as total gradient $\quad \frac{\partial}{\partial x^\mu}\left(F^{\alpha\beta}F_\beta{}^\mu\right) - \underbrace{F_\beta{}^\mu\frac{\partial F^{\alpha\beta}}{\partial x^\mu}}_{F_{\beta\mu}\frac{\partial F^{\alpha\beta}}{\partial x_\mu}}$

$$\frac{\partial T^{\alpha\beta}}{\partial x^\beta} = \frac{1}{\mu_0}\frac{\partial}{\partial x^\mu}\left(F^{\alpha\beta}F_\beta{}^\mu\right) - \frac{1}{\mu_0}F_{\beta\mu}\frac{\partial F^{\alpha\beta}}{\partial x_\mu}.$$

Faraday's law: $\dfrac{\partial F^{\alpha\beta}}{\partial x_\mu} + \dfrac{\partial F^{\mu\alpha}}{\partial x_\beta} + \dfrac{\partial F^{\beta\mu}}{\partial x_\alpha} = 0$

(this contains $\vec{\nabla}\cdot\vec{B} = 0$

$\vec{\nabla}\times\vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$)

Therefore $-F_{\beta\mu}\dfrac{\partial F^{\alpha\beta}}{\partial x_\mu} = -\dfrac{1}{2}F_{\beta\mu}\dfrac{\partial F^{\alpha\beta}}{\partial x_\mu} - \dfrac{1}{2}F_{\beta\mu}\left(-\dfrac{\partial F^{\mu\alpha}}{\partial x_\beta} - \dfrac{\partial F^{\beta\mu}}{\partial x_\alpha}\right)$

$(F^{\alpha\beta} = -F^{\beta\alpha})$ $= \underbrace{\dfrac{1}{2}\underset{\substack{\uparrow \\ \text{anti symmetric}}}{F_{\beta\mu}}\underbrace{\left(\dfrac{\partial F^{\beta\alpha}}{\partial x_\mu} + \dfrac{\partial F^{\mu\alpha}}{\partial x_\beta}\right)}_{\text{symmetric in } \mu,\beta}}_{\phi} + \dfrac{1}{2}F_{\beta\mu}\dfrac{\partial F^{\beta\mu}}{\partial x_\alpha} =$

$= \dfrac{1}{4}\dfrac{\partial}{\partial x_\alpha}\left(F_{\beta\mu}F^{\beta\mu}\right)$

$= \dfrac{1}{4}F_{\beta\mu}\dfrac{\partial F^{\beta\mu}}{\partial x_\alpha} + \underbrace{\dfrac{1}{4}F^{\beta\mu}\dfrac{\partial F_{\beta\mu}}{\partial x_\alpha}}_{\frac{1}{4}F_{\beta\mu}\frac{\partial F^{\beta\mu}}{\partial x_\alpha}}$

$$\Rightarrow \quad \frac{\partial T^{\alpha\beta}}{\partial x^\beta} = \frac{1}{\mu_0}\frac{\partial}{\partial x^\mu}\left(F^{\alpha\beta}F_\beta{}^\mu\right) + \frac{1}{4\mu_0}\frac{\partial}{\partial x_\alpha}\left(F_{\beta\mu}F^{\beta\mu}\right)$$

(can change $\mu \leftrightarrow \beta$ here)

Re-arrange indices: $\dfrac{\partial}{\partial x_\alpha}\left(F_{\beta\mu}F^{\beta\mu}\right) = \dfrac{\partial}{\partial x_\alpha}\left(F_{\mu\beta}F^{\mu\beta}\right) =$

$= \dfrac{\partial}{\partial x_\alpha}\left(F_{\mu\nu}F^{\mu\nu}\right) = \dfrac{\partial}{\partial x^\beta}g^{\alpha\beta}F_{\mu\nu}F^{\mu\nu}$

$g_{ik}A^k = A_i$, $g^{ik}A_k = A^i$, $A^iA_i = g_{ik}A^kA^i$

$\dfrac{\partial}{\partial x_\alpha} = (\text{contravariant}) \Rightarrow \hat{f}^\alpha$

$\dfrac{\partial}{\partial x^\beta} = (\text{covariant}) = \hat{f}_\beta \Rightarrow \hat{f}^\alpha = g^{\alpha\beta}\hat{f}_\beta \Rightarrow \dfrac{\partial}{\partial x_\alpha} = g^{\alpha\beta}\dfrac{\partial}{\partial x^\beta}$

so that $d\varphi = \hat{f}_\beta(\varphi)\,dx^\beta$ is scalar.

$$\Rightarrow \quad \frac{\partial}{\partial x^\beta}\left(T^{\alpha\beta}\right) = \frac{1}{\mu_0}\frac{\partial}{\partial x^\beta}\left(F^{\alpha\mu}F_\mu{}^\beta + \frac{1}{4}g^{\alpha\beta}F_{\mu\nu}F^{\mu\nu}\right)$$

$$F_\mu{}^\beta = g_{\mu\nu}F^{\nu\beta}$$

$$T^{\alpha\beta}_{field} = \frac{1}{\mu_0}\left(F^{\alpha\mu}g_{\mu\nu}F^{\nu\beta} + \frac{1}{4}g^{\alpha\beta}F_{\mu\nu}F^{\mu\nu}\right)$$

$$T^{\beta\alpha}_{field} = \frac{1}{\mu_0}\left(\underbrace{F^{\beta\mu}g_{\mu\nu}F^{\nu\alpha}}_{\underbrace{F^{\beta\nu}g_{\nu\mu}F^{\mu\alpha}}_{F^{\alpha\mu}g_{\mu\nu}F^{\nu\beta}}} + \frac{1}{4}\underbrace{g^{\beta\alpha}}_{g^{\alpha\beta}}F_{\mu\nu}F^{\mu\nu}\right)$$

$$\Rightarrow \quad T^{\alpha\beta} = T^{\beta\alpha}.$$

$$T^{00} = \frac{1}{\mu_0}\left(F^{0\mu}g_{\mu\nu}F^{\nu 0} + \frac{1}{4}\underbrace{g^{00}}_{=1}\underbrace{F_{\mu\nu}F^{\mu\nu}}_{2\left(\vec{B}^2 - \frac{1}{c^2}\vec{E}^2\right)}\right) = \frac{1}{\mu_0}\left(\frac{1}{2c^2}\vec{E}^2 + \frac{1}{2}\vec{B}^2\right) = \frac{\varepsilon_0}{2}\vec{E}^2 + \frac{1}{2\mu_0}\vec{B}^2.$$

$$F^{0\mu}g_{\mu\nu}F^{\nu 0} = \underbrace{F^{01}}_{-\frac{E_x}{c}}\underbrace{g_{11}}_{-1}\underbrace{F^{10}}_{\frac{E_x}{c}} + F^{02}g_{22}F^{20} + F^{03}g_{33}F^{30} = \frac{\vec{E}^2}{c^2}$$

$$F^{\alpha\beta} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

$$T^{01} = \frac{1}{\mu_0} F^{0\mu} g_{\mu\nu} F^{\nu 1} = \frac{1}{\mu_0}\left(F^{01} \cancel{g_{11} F^{11}} + F^{02} \underset{-1}{\underset{\parallel}{g_{22}}} F^{21} + F^{03} \underset{-1}{\underset{\parallel}{g_{33}}} F^{31}\right)$$

$$= -\frac{1}{\mu_0}\left(\left(-\frac{E_y}{c}\right) B_z + \left(-\frac{E_z}{c}\right)(-B_y)\right)$$

$$= \frac{1}{\mu_0 c}\left(E_y B_z - E_z B_y\right) = \frac{1}{c} S_x$$

Poynting vector $\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$

Clearly $T^{02} = \frac{1}{c} S_y$, $T^{03} = \frac{1}{c} S_z$.

<u>Space components</u>

$$T^{11} = \frac{1}{\mu_0}\left(F^{1\mu} g_{\mu\nu} F^{\nu 1} - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}\right) = \left(g^{11} = -1\right)$$

$$= \frac{1}{\mu_0}\left(F^{10} F^{01} - F^{12} F^{21} - F^{13} F^{31} - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}\right)$$

$$= \frac{1}{\mu_0}\left(-\frac{E_x^2}{c^2} + B_z^2 + B_y^2 - \frac{1}{2}\left(\vec{B}^2 - \frac{\vec{E}^2}{c^2}\right)\right)$$

$$= \frac{1}{2\mu_0 c^2}\left(\vec{E}^2 - 2E_x^2\right) + \frac{1}{2\mu_0}\left(2(B_y^2 + B_z^2) - \vec{B}^2\right)$$

$$= \frac{\varepsilon_0}{2}\left(E_y^2 + E_z^2 - E_x^2\right) + \frac{1}{2\mu_0}\left(B_y^2 + B_z^2 - B_x^2\right)$$

$$T^{12} = \frac{1}{\mu_0} F^{1\mu} g_{\mu\nu} F^{\nu 2} = \frac{1}{\mu_0}\left(F^{10} F^{02} - F^{13} F^{32}\right) =$$

$$= \frac{1}{\mu_0}\left(-\frac{E_x E_y}{c^2} - B_y B_x\right) = -\varepsilon_0 E_x E_y - \frac{B_x B_y}{\mu_0}$$

$$\Rightarrow T^{ij} = \varepsilon_0\left(\frac{\delta^{ij}}{2} \vec{E}^2 - E^i E^j\right) + \frac{1}{\mu_0}\left(\frac{\delta^{ij}}{2} \vec{B}^2 - B^i B^j\right) = -\sigma^{ij}$$

The meaning? Go back to

$$\frac{\partial T^{\alpha\beta}_{field}}{\partial x^\beta} = -F^{\alpha\beta} j_\beta \Rightarrow \text{spatial part } \frac{\partial T^{i\beta}}{\partial x^\beta} = -F^{i\mu} j_\mu = -f^i$$

here $f^i = \left(\rho \vec{E} + \vec{j} \times \vec{B}\right)^i$

$$\frac{\partial}{\partial x^\beta} T^{i\beta} = \frac{1}{c}\frac{\partial}{\partial t} T^{i0} + \frac{\partial}{\partial x^j} T^{ij} = -f^i$$

but $T^{i0} = T^{0i} = \frac{1}{c} S^i$

$$\frac{1}{c^2}\frac{\partial S^i}{\partial t} + \frac{\partial T^{ij}}{\partial x^j} = -f^i$$

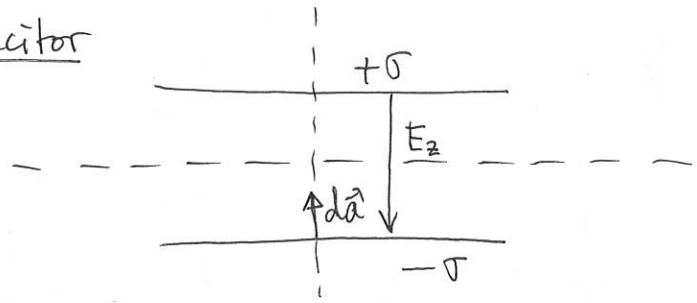
Assume constant EM field $\Rightarrow \frac{\partial S^i}{\partial t} = 0.$

$$\Rightarrow \frac{\partial \sigma^{ij}}{\partial x^j} = f^i \qquad \left(\sigma^{ij} = -T^{ij}\right)$$

$$\Rightarrow \underset{\substack{\uparrow \\ \text{net force}}}{F^i} = \int_V dV f^i = \int_V dV \frac{\partial \sigma^{ij}}{\partial x^j} = \underbrace{\oint_S da_j \, \sigma^{ij}}_{\substack{\text{net force} \\ \text{applied to the} \\ \underline{\text{surface}} \text{ by EM field}}}$$

This is non-trivial:
we only need to know EM field on the surface in order to find the total force produced by EM field irrespective of the behavior of the field inside and properties of the material.

Capacitor



$$E_z = -\frac{\sigma}{\varepsilon_0}, \quad E_{x,y} = 0$$

$$\vec{B} = 0$$

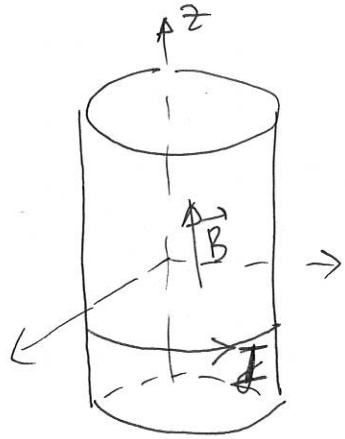
Maxwell's stress tensor

$$\sigma^{ij} = \varepsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} \vec{E}^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} \delta_{ij} \vec{B}^2 \right)$$

$$= \varepsilon_0 \begin{pmatrix} -\frac{1}{2} E_z^2 & 0 & 0 \\ 0 & -\frac{1}{2} E_z^2 & 0 \\ 0 & 0 & \frac{1}{2} E_z^2 \end{pmatrix} = \frac{1}{2} \varepsilon_0 \left(\frac{\sigma}{\varepsilon_0} \right)^2 \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

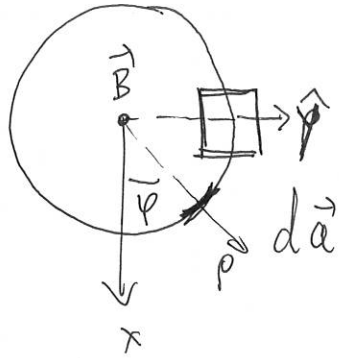
$$F^i = \int_V dV f^i = \oint_S da^j \, \sigma^{ij} = \oint_S da_z \, \sigma_{zz} = A \cdot \frac{\sigma^2}{2\varepsilon_0}$$

Solenoid



$$\vec{B} = (0, 0, B)$$

$$\sigma_{ij} = \frac{1}{2\mu_0} B^2 \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



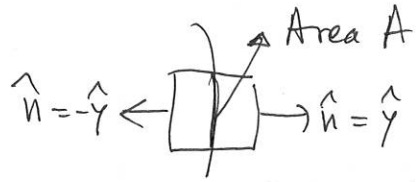
$$da = R \, d\varphi$$

$$d\vec{a} = da_x \cos\varphi \, \hat{x} + da_y \sin\varphi \, \hat{y}$$

$$F_x = \oint \sigma_{xx}\, da_x + \underset{\overset{\shortparallel}{0}}{\sigma_{xy}}\, da_y + \underset{\overset{\shortparallel}{0}}{\sigma_{xz}}\, da_z = \oint \sigma_{xx}\, R\, d\varphi \cos\varphi$$

$$F_y = \oint \sigma_{yy}\, da_y = \oint \sigma_{yy}\, R\, d\varphi \sin\varphi$$

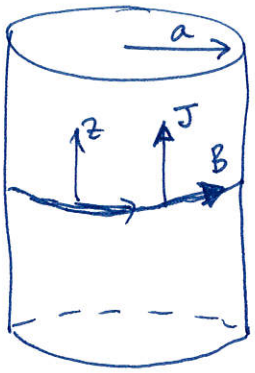
The normal



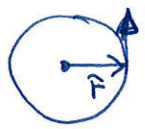
$$F_y = \oint (-dA) \cdot \left(-\frac{B^2}{2\mu_0} \right) = A \cdot \frac{B^2}{2\mu_0}$$

$\Rightarrow$ magnetic field pressure $P = \dfrac{B^2}{2\mu_0}$.

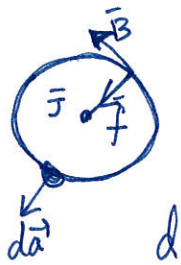
Cylindrical wire with uniform current $\vec{J}$.



$$\vec{B}(r) = \begin{cases} \dfrac{\mu_0 I}{2\pi r} & \text{for } r \geq a \\ 0 & r < a \end{cases}$$



$$\vec{B} = \frac{\mu_0 I}{2\pi r}\hat{\phi} = \frac{\mu_0 I}{2\pi r}\left(-\sin\varphi\,\hat{x} + \cos\varphi\,\hat{y}\right) \quad \begin{pmatrix} B_\rho = 0 \\ B_z = 0 \end{pmatrix}$$



$$\vec{J}\times\vec{B} \sim \hat{z}\times\hat{\phi} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & 1 \\ -\sin\varphi & \cos\varphi & 0 \end{vmatrix} = -\cos\varphi\,\hat{x} - \hat{y}\sin\varphi = -\hat{\rho}$$

$$d\vec{a} = dz\, r\, d\varphi\, \hat{\rho} = dz\, r\, d\varphi\left(\cos\varphi\,\hat{x} + \sin\varphi\,\hat{y}\right)$$

$$F_\rho = \int \sigma_{\rho\rho}\, da_\rho + \sigma_{\varphi\rho}\, da_\rho$$

$$\sigma_{\rho\rho} = \frac{1}{\mu_0} B_\rho^2 - \frac{1}{2\mu_0}\left(B_\rho^2 + B_\varphi^2 + B_z^2\right) = -\frac{B_\varphi^2}{2\mu_0}$$

$$\sigma_{\varphi\rho} = \sigma_{z\rho} = 0$$

$$F_\rho = -\frac{1}{2\mu_0}\int \left(\frac{\mu_0 I}{2\pi a}\right)^2 dz\, a\, d\varphi =$$

$$= -\frac{1}{2\mu_0}\left(\frac{\mu_0 I}{2\pi a}\right)^2 L\, a\, 2\pi = -\frac{\mu_0 I^2 (La)}{4\pi a^2}$$

(sign of the force?)

(The energy argument does not work because total energy $\int dr\, r \left(\frac{I}{r}\right)^2 \sim \int_a^\infty \frac{dr}{r}$ is divergent) $U \sim \ln\frac{R}{a}$, $\frac{\partial U}{\partial a} = -\frac{1}{a}$

$U_1 - U_2 = \ln\frac{a_2}{a_1} > 0$

$\left(\frac{a_2}{a_1}\right)$ $U_2 < U_1$!

$$\vec{F} = \int dV \left(\rho \vec{E} + \vec{J} \times \vec{B} \right) = \int dV \vec{f} \text{ force on a volume of charge.}$$

$$\rho = \varepsilon_0 \vec{\nabla} \cdot \vec{E} \quad , \quad \vec{J} = \frac{1}{\mu_0} \left(\vec{\nabla} \times \vec{B} - \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

$$\vec{f} = \varepsilon_0 \vec{E} (\vec{\nabla} \cdot \vec{E}) + \frac{1}{\mu_0} \left(\vec{\nabla} \times \vec{B} - \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \times \vec{B}$$

$$\frac{\partial}{\partial t} (\vec{E} \times \vec{B}) = \frac{\partial \vec{E}}{\partial t} \times \vec{B} + \vec{E} \times \underbrace{\frac{\partial \vec{B}}{\partial t}}_{-\vec{\nabla} \times \vec{E}} \Rightarrow \frac{\partial \vec{E}}{\partial t} \times \vec{B} = \frac{\partial}{\partial t} (\vec{E} \times \vec{B}) + \vec{E} \times (\vec{\nabla} \times \vec{E})$$

$$\vec{f} = \varepsilon_0 \vec{E} (\vec{\nabla} \cdot \vec{E}) - \varepsilon_0 \vec{E} \times (\vec{\nabla} \times \vec{E}) + \frac{1}{\mu_0} (\vec{\nabla} \times \vec{B}) \times \vec{B} - \varepsilon_0 \frac{\partial}{\partial t} (\vec{E} \times \vec{B})$$

$$= \varepsilon_0 \left[\vec{E} (\vec{\nabla} \cdot \vec{E}) - \vec{E} \times (\vec{\nabla} \times \vec{E}) \right]$$

$$+ \frac{1}{\mu_0} \left[\vec{B} (\vec{\nabla} \cdot \vec{B}) - \vec{B} \times (\vec{\nabla} \times \vec{B}) \right]$$

$$- \varepsilon_0 \frac{\partial}{\partial t} \underbrace{(\vec{E} \times \vec{B})}_{\mu_0 \vec{S} = \text{Poynting vector}}$$

add $\vec{B} \underbrace{(\vec{\nabla} \cdot \vec{B})}_{= 0 \,!} = 0$

because it is just zero!

Show that $\left\{ \vec{E} (\vec{\nabla} \cdot \vec{E}) - \vec{E} \times (\vec{\nabla} \times \vec{E}) \right\}_j = \frac{\partial}{\partial x_i} \left(E_i E_j - \frac{1}{2} \delta_{ij} \vec{E} \cdot \vec{E} \right)$

(sum over i here!)

$i = 1, 2, 3 \; (x, y, z)$

$$f_j^{(e)} = E_j \frac{\partial}{\partial x_i} E_i - \varepsilon_{jik} \varepsilon_{klm} E_i \frac{\partial}{\partial x_l} E_m =$$

$$\vec{E} \times \vec{D} \Rightarrow \varepsilon_{jik} E_i D_k$$

$$\vec{D} = \vec{\nabla} \times \vec{E} \Rightarrow \varepsilon_{klm} \frac{\partial}{\partial x_l} E_m$$

$$\varepsilon_{kji} \varepsilon_{klm} = \delta_{jl} \delta_{im} - \delta_{jm} \delta_{il}$$

$$= E_j \frac{\partial}{\partial x_i} E_i - \delta_{jl} \delta_{im} E_i \frac{\partial}{\partial x_l} E_m + \delta_{jm} \delta_{il} E_i \frac{\partial}{\partial x_l} E_m$$

$$= E_j \frac{\partial E_i}{\partial x_i} - E_i \frac{\partial}{\partial x_j} E_i + E_i \frac{\partial}{\partial x_i} E_j =$$

$$= \frac{\partial}{\partial x_i}\left(E_j E_i\right) - \frac{1}{2}\frac{\partial}{\partial x_j}\left(E_i E_i\right) = \frac{\partial}{\partial x_i}\left(E_i E_j - \frac{1}{2}\delta_{ij}\vec{E}^2\right)$$

$$\Rightarrow \quad f_j = \varepsilon_0 \frac{\partial}{\partial x_i}\left(E_i E_j - \frac{1}{2}\delta_{ij}\vec{E}^2\right) + \frac{1}{\mu_0}\frac{\partial}{\partial x_i}\left(B_i B_j - \frac{1}{2}\delta_{ij}\vec{B}^2\right)$$

$$- \varepsilon_0 \mu_0 \frac{\partial}{\partial t}\left(\vec{E} \times \vec{B}\right)_j$$

$$\varepsilon_{jil} E_i B_l$$

Divergence of tensor $\overleftrightarrow{\sigma}$ gives vector $\vec{f}$.